

[Quiz—3] Spring 2024: CSE 309: Theory of Automata

— SOLUTION —

Tuesday, April 2, 2024. Total Marks: 10, Duration: 15 minutes

Question 1

[10 points]

Design a Turing machine for the following language L over the alphabet $\Sigma = \{0, 1\}$:

$$L = \{uvu^{\text{rev}} : u, v \in \Sigma^*\}.$$

Here w^{rev} denotes the reverse of the string w , e.g., if $w = 00110$, then $w^{\text{rev}} = 01100$.

[**Note:** You may provide your solution as a description or represent it graphically.]

Solution: The language L actually has all possible strings over the alphabet Σ . So, the following Turing machine M for L should accept all strings. The Turing machine can be designed as follows:

$M =$ “On input string w :

1. Repeat the following steps until the tape is empty:
 - (a) If the current symbol is 0 or 1, then erase it and move the tape head to the right.
 - (b) If the current symbol is $\sqcup$, then halt and accept.